

EVA Allo overview

Remarks

- PE + router are implemented as a single kernel within a `df.region()`
 - o EVA specific router version: turning is core-mediated
 - o Fusing prevented addition of additional streams
- Two networks (implemented using Allo Streams)
 - o Systolic
 - o Router (+ credit links)
- Bubble model/always-send model is used (for systolic/router/credit)
 - o We send and receive on ALL links every cycle
 - o If there is no valid value to transmit on a specific link in a cycle, a bubble (invalid value/packet) is transmitted
 - o The bubble model is needed because we only have blocking reads in Allo. However, using blocking reads would lead to stalling the overall grid
- Router network: Bubble model + credits
 - o Each PE holds a credit counter for each direction
 - o High-level idea: only send a packet downstream if it can be stored -> no packet loss
 - o If packet was sent downstream -> decrease credit counter for this direction
 - o If downstream frees a register slot: sent +1 upstream -> upstream increases credit counter -> can send a packet downstream only if positive credit balance
 - o EVA specific: packet turns are core mediated!
- Systolic network: Bubble model
- Sync registers (EVA-specific):
 - o Configure a DRF register as 2-slot FIFO (instead of a normal register)
 - o If DRF register is used in `sync_mode` a value cannot just be overwritten but must be 'consumed'
 - o Used in FFT

Variable	Description
drf: Ty[DRF_DEPTH]	Data register file
drf_full: int32[DRF_DEPTH]	Per slot full bit
dsmask: int32	DRF sync mask (which of the DRF slots is a sync register)
Crv_valid: int32	Packet was delivered from router to core
Crv_data: Ty	Payload that was delivered
Crv_addr : int32	DRF slot of delivered packet
Crv_mode: int32	Data or programming mode
Crv_raw : int32	Delivered payload
Csd_vld: int32	Core wants to send packet
Csd_pkt: Pkt	Packet
Csd_dir: int32	Injection direction
Hold_v: Ty[4,2]	Per direction: registers to hold systolic inputs
Hold_cnt: int32[4]	How many values are occupied
Rbuf: Pkt[4, BUF_DEPTH]	Per direction: registers to hold router packets
Rbcnt: int32[4]	How many values are occupied
Rcred: int32[4]	Credits held per direction = free buffer downstream node
Cfg_isz: int32	How many instructions are used for this kernel (1-8)
Cfg_itsz: int32	Iteration size: how many times to loop kernel
Fetch_en: int32	Start bit
Instr_cnt: int32	Current position within kernel (0...cfg_isz)
Iter_cnt: int32	How many kernel loops have completed
NSTEP	Cycles the always-fire model loops -> set it to high value

Hd: Pkt[4] Hvld: int32[4] Hit: int32[4] Axis: int32[4]	Per direction: oldest packet in rbuf Valid if there is a packet 1 if packet is destined for this core West, East = horizontal links -> match col_id. North, Sout = vertival links -> match row_id
O_out: Pkt[4] Pop: int32[4] Inj_done: int32 Idir: int32	Per direction: packet scheduled to be sent Which buffer head should be dequeued Got core's inject sent Direction core wants to sent to
Ret: int32[4]	Was a slot freed this cycle?
Pc: int32 Instr: int32 grant	Program counter Current instruction Instruction could execute (all inputs valid -> pc can advance otherwise: stall on missing operand)

Overview of variables used in eva.py (not every variable is listed)

Ty: data type, e.g. float16

